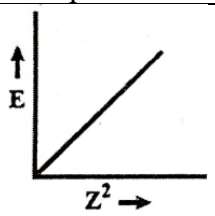
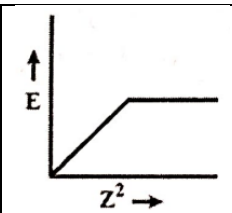
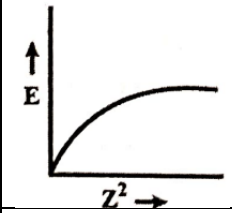
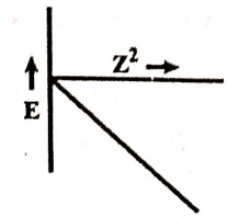
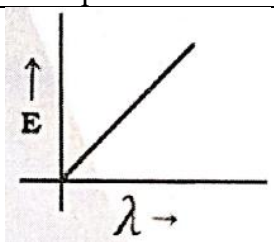
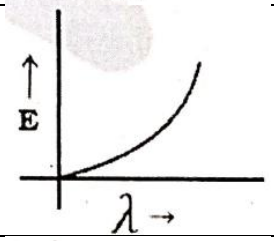
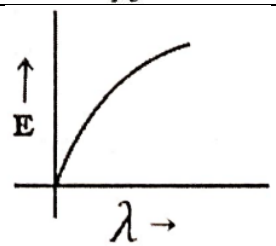
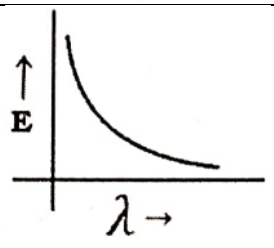
Question	Who discovered the electrons?
Type	multiple choice
Option	E. Goldstein
Option	J. Chadwick
Option	J.J. Thomson
Option	Neils Bohr
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of specific charge of a proton and an α-particle is:
Type	multiple choice
Option	$2:1$
Option	$1:2$
Option	$1:4$
Option	$1:1$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The radius of nucleus 1327A*l* will be:
Type	multiple choice
Option	$2\times10^{-15}\,\mathrm{m}$
Option	$2.7\times10^{-15}\,\mathrm{m}$
Option	$10.8\times10^{-15}\,\mathrm{m}$
Option	$4\times10^{-15}\,\mathrm{m}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Rutherford's α-particle scattering experiment proved that atom has:
Type	multiple choice
Option	Electrons
Option	Neutrons
Option	Nucleus
Option	Orbitals
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A and B are two elements which have same atomic weight and are having atomic number 27 and 30 respectively. If the atomic weight of A is 57 then number of neutron in B is:
Type	multiple choice
Option	27
Option	33
Option	30
Option	40
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	The radius of a shell for H-atom is 4.761A°. The value of n is:
Type	multiple choice
Option	3
Option	9
Option	5
Option	4
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	For Li^{+2} ion, $r_2 : r_2$ will be
Type	multiple choice
Option	$9:25$
Option	$4:25$
Option	$25:4$
Option	$25:9$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Angular momentum in second Bohr orbit of H-atom is x. The find out angular momentum in 1st excited state of Li^{+2} ion:
Type	multiple choice
Option	$3x$
Option	$9x$
Option	$\frac{x}{2}$
Option	x
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Potential energy is -27.2 eV in second orbit of He+, then calculate double of total energy in first excited state of hydrogen atom:
Type	multiple choice
Option	-13.6 eV
Option	-54.4 eV
Option	-6.8 eV
Option	-27.2 eV
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The energy of H-atom in nth orbit is E_n, then energy in nth orbit of singly ionized helium ion will be:
Type	multiple choice
Option	$4E_n$
Option	$E_n / 4$
Option	$2E_n$
Option	$E_n / 2$

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The energy of one mole photons of radiation having frequency 5.01×10^{14} Hz is:
Type	$160\ KJ\ mol^{-1}$
Option	$180\ KJ\ mol^{-1}$
Option	$200\ KJ\ mol^{-1}$
Option	$220\ KJ\ mol^{-1}$
Option	3
Answer	
Solution	
Positive Marks	
Negative Marks	
Question	What is the energy content per photon (J) for light of frequency 4.2×10^{14} ?
Type	multiple choice
Option	2.8×10^{-21}
Option	2.5×10^{-19}
Option	2.8×10^{-19}
Option	2.5×10^{-18}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Wavelength for high energy EMR transition in H-atom is 91 nm. What energy is needed for this transition?
Type	multiple choice
Option	1.36 eV
Option	1240 eV
Option	13 eV
Option	13.6 eV
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	$E_n = -313.6 / n^2$. If the value of $E_n = -34.84$ then to which of the following value does 'n' correspond:
Type	multiple choice
Option	1
Option	2
Option	3
Option	4
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If the ionization energy of hydrogen is 313.8 kcal per mol, then the energy of the electron in 2nd excited state will be:

Type	multiple choice
Option	−113.2 kcal/mol
Option	−78.45 kcal/mol
Option	−313.8 kcal/mol
Option	−35 kcal/mol
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is not a correct order of energy for 1st, 2nd and 3rd orbit:
Type	multiple choice
Option	$E_1 > E_2 > E_3$
Option	$(PE)_1 < (PE)_2 < (PE)_3$
Option	$(KE)_1 > (KE)_2 > (KE)_3$
Option	Both (a) and (c)
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The radiation of low frequency will be emitted in which transition of hydrogen atom:
Type	multiple choice
Option	n = 1 to n = 4
Option	n = 2 to n = 5
Option	n = 3 to n = 1
Option	n = 5 to n = 2
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following electron transition will require the largest amount of energy in a hydrogen atom:
Type	multiple choice
Option	From n = 1 to n = 2
Option	From n = 2 to n = 3
Option	From n = 1 to n = ∞
Option	From n = 3 to n = 5
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The graphical representation of energy of electron and atomic number is:
Type	multiple choice
Option	

Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of energies of hydrogen atom for first and second excited state is:
Type	multiple choice
Option	$4/1$
Option	$1/4$
Option	$4/9$
Option	$9/4$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Assume that 2×10^{-17} J of light energy is needed by the interior of the human eye to see an object. How many photons of yellow light with λ = 595.2 nm are needed to generate this minimum energy?
Type	multiple choice
Option	6
Option	30
Option	45
Option	60
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between wavelengths i.e., λ_1 and λ_2 will be:
Type	multiple choice
Option	$\lambda_1 = \lambda_2$
Option	$\lambda_1 = 2\lambda_2$
Option	$\lambda_1 = 4\lambda_2$
Option	$\lambda_1 = \frac{1}{2}\lambda_2$

Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at:
Type	multiple choice
Option	743 nm
Option	518 nm
Option	1035 nm
Option	325 nm
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which graph shows how the energy E of a photon of light is related to its wavelengths (λ)?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If an electron is travelling at 200 m/s within 1 m/s uncertainty, what is the theoretical uncertainty in its position in ovum (micrometer)?
Type	multiple choice
Option	14.5
Option	29
Option	58

Option	114
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The uncertainty in position of an electron & helium atom are same. If the uncertainty in momentum for the electron is $32\times10^{5}\ g\ cm\ s^{-1}$, then the uncertainty in momentum of helium atom will be:
Type	multiple choice
Option	$32\times10^{5}\ g\ cm\ s^{-1}$
Option	$16\times10^{5}\ g\ cm\ s^{-1}$
Option	$8\times10^{5}\ g\ cm\ s^{-1}$
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4Å, the circumference of the orbit will be:
Type	multiple choice
Option	4Å
Option	4 nm
Option	16Å
Option	16 nm
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the ratio of the de-Broglie wavelength for electrons accelerated through 200 volts and 50 volts:
Type	multiple choice
Option	$1:2$
Option	$2:1$
Option	$3:10$
Option	$10:3$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An electron has a kinetic energy of 2.8×10^{-23} J. De-Broglie wavelength will be nearly: $\left(m_e = 9.1\times10^{-31} kg\right)$
Type	multiple choice
Option	$9.28\times10^{-31} m$
Option	$9.28\times10^{-7} m$
Option	$9.28\times10^{-8} m$
Option	$9.28\times10^{-10} m$
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	An electron travels with a velocity of $x\ ms^{-1}$. For a proton to have the same de-Broglie wavelength, the velocity will be approximately:
Type	multiple choice
Option	$\frac{1840}{x}$
Option	$\frac{x}{1840}$
Option	$1840x$
Option	x
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The following quantum number are 9. possible for how many orbitals $n = 3,\ l = 2,\ m = +2$?
Type	multiple choice
Option	1
Option	2
Option	3
Option	4
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Nodal plane is found in which orbital:
Type	multiple choice
Option	$n = 2,\ l = 0$
Option	$n = 3,\ l = 0$
Option	$n = 2,\ l = 1$
Option	$n = 1,\ l = 0$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is correct for a 4d-electron?
Type	multiple choice
Option	$n = 4, l = 2, s = +\frac{1}{2}$
Option	$n = 4, l = 2, s = 0$
Option	$n = 4, l = 3, s = 0$
Option	$n = 4, l = 3, s = +\frac{1}{2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which statement is not correct for $n = 5,\ m = 2$?
Type	multiple choice

Option	$l=4$
Option	$l=0,1,2,3; s=+\frac{1}{-}$
Option	$l=3$
Option	$l=2,3,4$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The maximum number of electrons in a $p-$orbital with $n=6$ and $m=0$ can be:
Type	multiple choice
Option	14
Option	6
Option	2
Option	10
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	An electron is in one of 4d orbital. Which of the following quantum number value is not possible?
Type	multiple choice
Option	$n=4$
Option	$l=1$
Option	$m=1$
Option	$m=2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of the minimum frequency of Lyman & Blamer series will be:
Type	multiple choice
Option	1.25
Option	0.25
Option	5.4
Option	10
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of wavelengths of first line of Lyman series in Li^{+2} and first line of Lyman series in deuterium $(12H)$ is:
Type	multiple choice
Option	$1:9$
Option	$9:1$
Option	$1:4$
Option	$4:1$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	The first emission line in the H-atom spectrum in the Balmer series will have wave number:
Type	multiple choice
Option	$\frac{5R}{36}cm^{-1}$
Option	$\frac{3R}{4}cm^{-1}$
Option	$\frac{7R}{144}cm^{-1}$
Option	$\frac{9R}{400}cm^{-1}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If 9.9 eV energy is supplied to H atom, the number of spectral lines emitted is equal to:
Type	multiple choice
Option	0
Option	1
Option	2
Option	3
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The. frequency of photons emitted in Paschen series is given by $v = 3.29\times10^{15}(hz)\left[\frac{1}{3^2}-\frac{1}{n^2}\right]$. Value of n for photon having wavelength 1285 nm will be:
Type	multiple choice
Option	$n=3$
Option	$n=4$
Option	$n=5$
Option	$n=6$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In H-atom, electron transits from 6th orbit to 2nd orbit in multi-step. Then total spectral lines (without Balmer series) will be:
Type	multiple choice
Option	6
Option	10
Option	4
Option	0
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If $l=3$ then type and number of orbital is:

Type	multiple choice
Option	$3p,3$
Option	$4f,14$
Option	$5f,7$
Option	$3d,5$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which orbital has two angular nodal (d) m = 2 planes?
Type	multiple choice
Option	s
Option	p
Option	d
Option	f
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For $Na(Z=11)$ set of quantum number for last electron is:
Type	multiple choice
Option	$n=3, l=1, m=1, s=+\frac{1}{2}$
Option	$n=3, l=\mathrm{O}, m=\mathrm{O}, s=+\frac{1}{2}$
Option	$n=3, l=0, m=1, s=+\frac{1}{2}$
Option	$n=3, l=1, m=1, s=-\frac{1}{2}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1